

三代升级款数控降压电源彩屏款

型号：ZK-6522C

产品参数

产品名称：数控直流降压电源（彩屏款）	产品型号：ZK-6522C
输入电压：6.0-75.00V	输出电压：0.0-65.00V
输出电流：0-22A；	输出电压精度：±0.3% +3个字（可校准）
输出功率：0-1430W；	输出电流精度：±0.5% +3个字（可校准）
电压分辨率：0.01V	电流分辨率：0.01A
数据组存储：11组	硅胶按键：5个
屏幕尺寸：1.8寸升级大屏，36°29mm可视范围	蜂鸣器：有
散热风扇开启条件：温度>45℃或电流>2A或功率>30W	
产品尺寸：外壳：83x48x29.4mm（高度不含旋转编码器）；主板：124x71x42mm	
产品重量：主板净重210g；整套净重243g，整套含包装重296g	

保护机制：

输入欠压保护(LUP)：5.5-64V可调，出厂默认值5.5V
输出过压保护(OUP)：1-66V可调，出厂默认值66V
输出过流保护(OCP)：0.001-23A可调，出厂默认值23A
输出过功率保护(OPP)：1-1450W可调，出厂默认值1450W
过温保护(OTP)：30-99℃可调，出厂默认值60℃
超时保护(OHP)：1分钟-99小时59分钟，出厂默认关闭
超容量保护(OAH)：0.001-9999Ah，出厂默认关闭
超能量保护(OWH)：0.001-4000KWh，出厂默认关闭

注意事项：

1. 不可以输入交流电，更不能接220V市电。2. 输入电压最高允许75V，不能超压输入或者正负极接反。
3. 输出不能完全保证防倒灌，接电池时要保证电源模块输出电压始终高于电池电压，或在输出正极串联二极管单向导通，防止电池电压倒灌烧毁电源。
4. 输出端接电池时注意正负极不要接反。5. 产品为降压模块，最大可输出电压=输入电压*0.95-1V。

按键功能

短按：调出数据组设置

短按：电压设置

长按：OVP设置

短按：电流设置

长按：OCP设置

短按：确定

短按：选择设置位

长按：打开/取消锁定功能

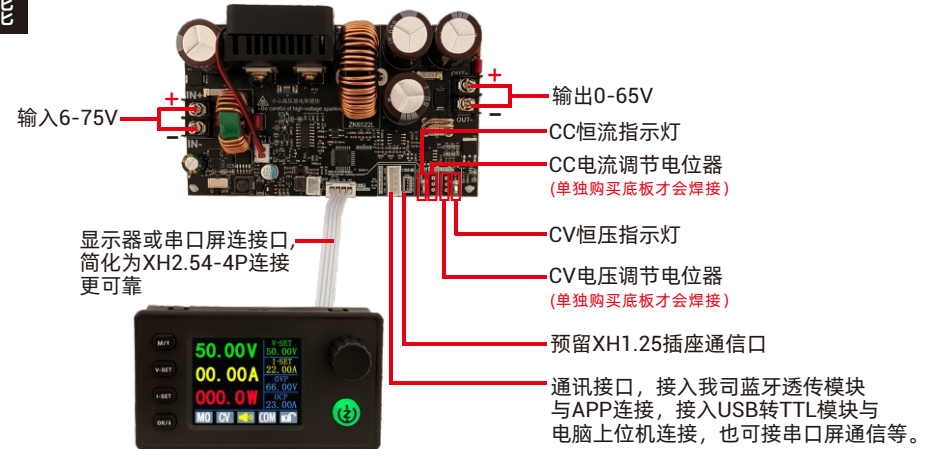
左旋转：切换界面显示；
设置时数字减

右旋转：切换界面显示；
设置时数字加

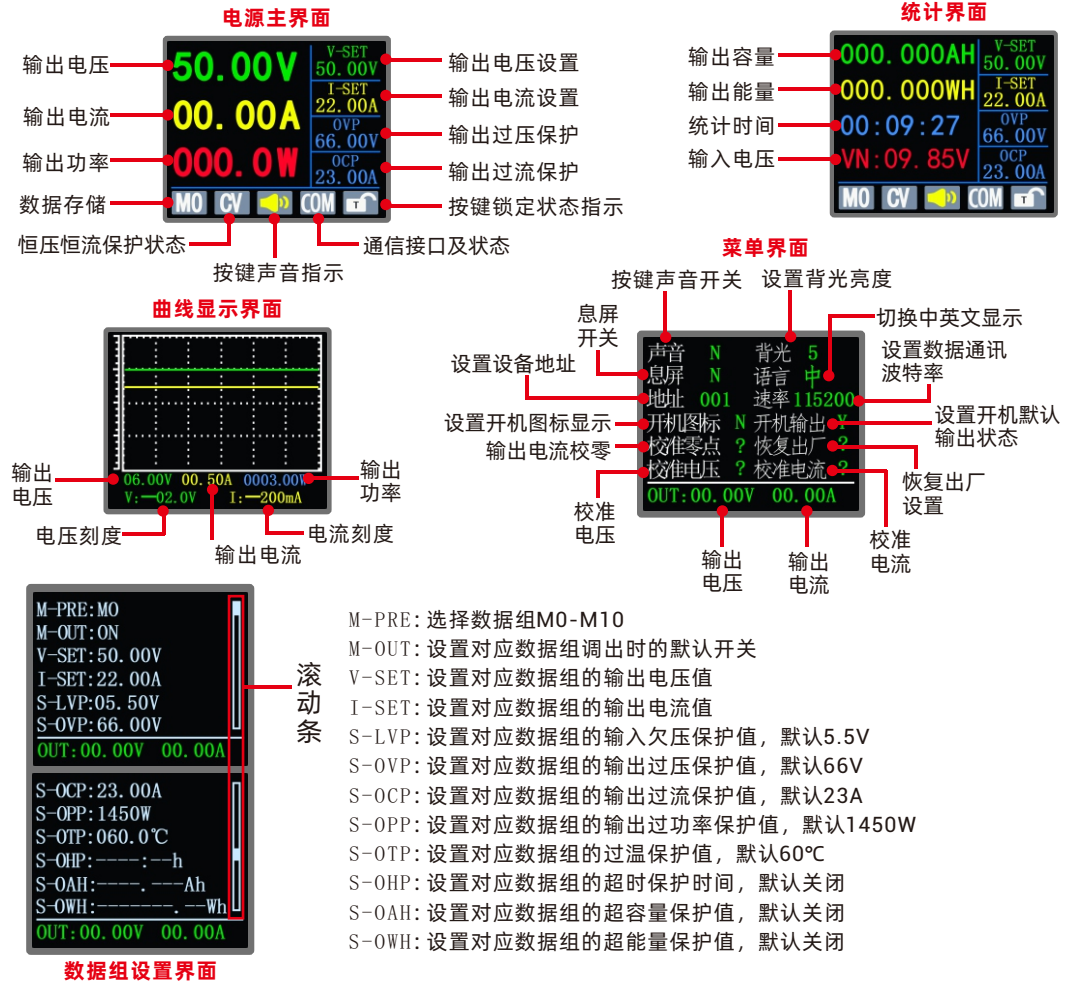
短按：关闭/打开输出

长按：休眠关机

产品功能



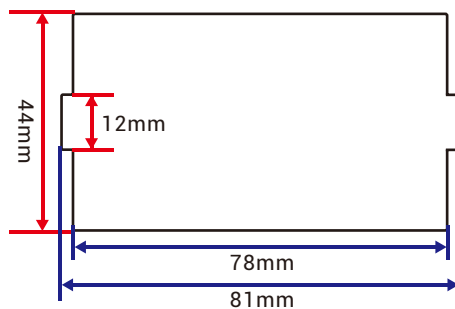
显示界面介绍



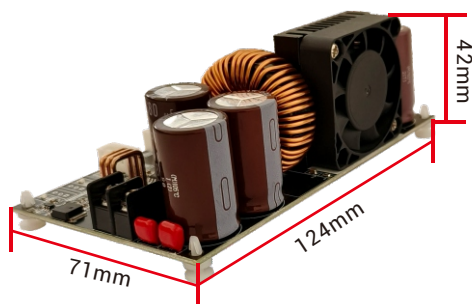
外壳尺寸



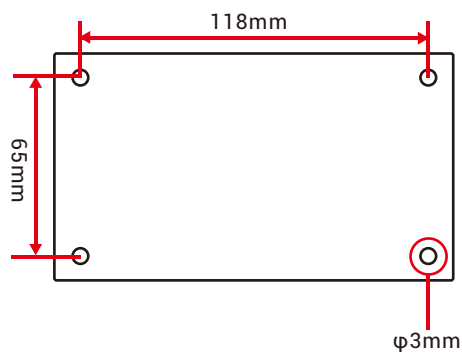
建议开口尺寸



底板尺寸

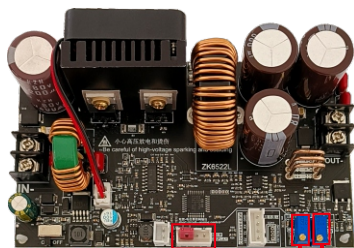


底板安装孔尺寸



底板可独立使用

输出电流0-22A可调，输出电压0-65V可调

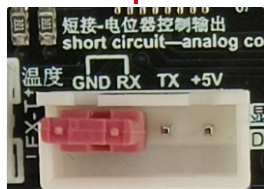


模拟电位器控制
时需短接RX GND,
否则为数字通信
控制

电压调节电位器

电流调节电位器

注意：ZK6522B单独主板售卖时，有焊接好电压、电流调节电位器，和插好跳线帽。如果是串口屏或APP/上位机控制，必须要拔掉跳线帽重新上电。

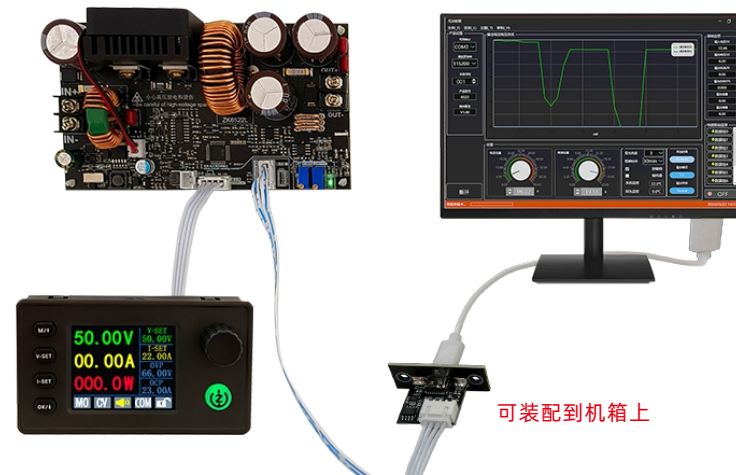


APP联机方法



- ① 电源主板通信接口插入ZK-BT蓝牙板，插入瞬间ZK-BT指示灯会闪一下。
- ② 打开无治智联APP和手机蓝牙，APP界面点击“添加设备”，会自动搜索到 Wuzhi Power，点击后自动连接。
- ③ 显示连接成功，点击完成。
- ④ 点击设备图标，进入正常操作界面，长按产品图标可更改名字或删除设备。

上位机联机方法



可装配到机箱上

- ① 电脑插入数据线连接ZK-U2T (usb转ttl模块)，ZK-U2T模块另一端4P线插入电源主板的通信口；
- ② 在我的电脑-设备管理器-端口中找到USB-SERIAL CH340对应的端口号COMx；
- ③ 电脑打开“无治智联”上位机，选择对应的端口号COMx，点击上位机左下角的“连接”按钮，自动联机。
- ④ 运行环境是 .NET Framework 4.8 如果有运行不了的可至 .net官网下载安装

使用说明

调用数据组



按 **M/↑** 调出数据组设置，旋转编码器切换数据组M0-M10；
按 **OK/↓** 确定并退出

设置电压电流



短按 **V-SET** 进入电压设置，短按旋转编码器移位，旋转调整大小。
设置好后，再按 **V-SET** 或 **OK/↓** 一下，退出设置。



短按 **I-SET** 进入电流设置，短按旋转编码器移位，旋转调整大小。
设置好后，再按 **I-SET** 或 **OK/↓** 一下，退出设置。

快捷设置OVP/OCF

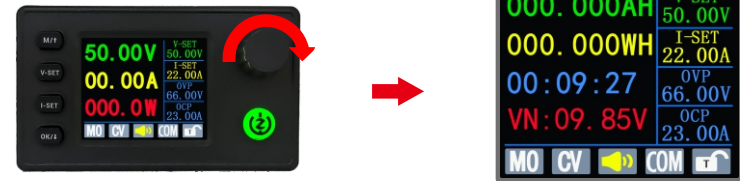


长按 **V-SET** 进入OVP设置，短按旋转编码器移位，旋转调整大小。
设置好后，再按 **V-SET** 或 **OK/↓** 一下，退出设置。

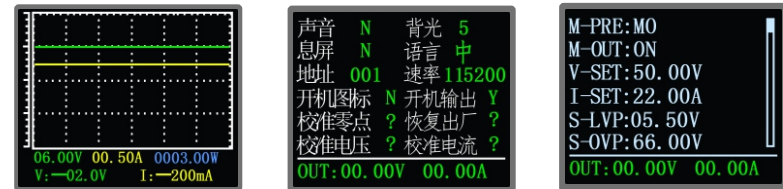


长按 **I-SET** 进入OCF设置，短按旋转编码器移位，旋转调整大小。
设置好后，再按 **I-SET** 或 **OK/↓** 一下，退出设置。

切换显示界面



旋转编码器切换界面显示



锁定功能



显示锁定符号

在正常显示界面，长按旋转编码器2秒，锁定设置的电压与电流，防止误操作；锁定后长按编码器2秒解锁。
锁定后电源键一样能够正常操作，确保随时可以断开电源。

参数设置操作介绍

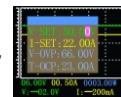
在正常界面，旋转编码器切换界面显示

曲线显示界面

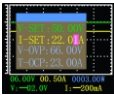


按 **OK/↓** 设置电压刻度（显示 **V: -02.0V**），旋转编码器切换大小，再按 **OK/↓** 切换到电流刻度（显示 **I: -200mA**），旋转编码器切换大小，再按 **OK/↓** 退出；短按旋转编码器暂停。

按 **M/↑** 调出数据组  旋转编码器切换数据组M0-M10；按 **OK/↓** 确定并退出

按 **V-SET** 调出设置界面，  可以设置电压、电流、过压保护、过流保护

按 **I-SET** 调出设置界面，



可以设置电压、电流、过压保护、过流保护

菜单界面

短按 **M/t** 选择设置选项



- 1. **声音 N**：旋转编码器打开/关闭
- 2. **屏显 N**：旋转编码器打开/关闭并设置息屏时间
- 3. **地址 001**：旋转编码器设置通信地址
- 4. **开机图标 N**：旋转编码器打开/关闭

- 5. **校准零点 ?**：短按 **OK/↓** 弹出校准零点对话框



短按 **M/t** 取消并退出，短按 **OK/↓** 确定并退出。

- 6. **校准电压 ?**：短按 **OK/↓** 弹出校准电压对话框



在对话框中短按 **M/t** 取消校准并退出。

校准步骤：

- 1. 用万用表测量现在输出端（输出端空载，测量电源端子）实际电压并输入。
 - 2. 短按 **OK/↓**，显示电压2。
 - 3. 用万用表再次测量现在输出端实际电压并输入。
 - 4. 短按 **OK/↓**，完成校准。
- 如果校准成功，会显示“校准成功”并退出，否则显示“校准失败”并退出。如果失败可尝试再次校准。

- 7. **背光 5**：旋转编码器调整背光亮度；0-5级
- 8. **语言 中**：旋转编码器切换中/英文
- 9. **速率 115200**：旋转编码器切换通信速率大小
- 10. **开机输出 Y**：旋转编码器调整开机默认输出开/关

- 11. **恢复出厂 ?**：短按 **OK/↓** 弹出校准恢复出厂对话框



短按 **M/t** 取消并退出，短按 **OK/↓** 确定并退出。

- 12. **校准电流 ?**：短按 **OK/↓** 弹出校准电流对话框



在对话框中短按 **M/t** 取消校准并退出。

校准步骤：

- 1. 用万用表电流档或用电子负载，短接输出两端测量现在实际电流并输入。
 - 2. 短按 **OK/↓**，显示电流2。
 - 3. 用万用表再次测量现在输出端实际电流并输入。
 - 4. 短按 **OK/↓**，完成校准。
- 如果校准成功，会显示“校准成功”并退出，否则显示“校准失败”并退出。如果失败可尝试再次校准。

数据组设置界面

短按 **M/t** 选择设置选项，设置完成后短按 **OK/↓** 确定并退出。



- 1. **M-PRE: M**：设置数据组
旋转编码器选择数据组
 - 2. **M-OUT: ON**：设置调出时默认开/关
旋转编码器设置调出时默认开/关
 - 3. **V-SET: 50.0 V**：设置输出电压
短按编码器移位，旋转编码器调整大小
 - 4. **I-SET: 22.0 A**：设置输出电流
短按编码器移位，旋转编码器调整大小
 - 5. **S-LVP: 05.5 V**：设置输入欠压保护
短按编码器移位，旋转编码器调整大小
 - 6. **S-OVP: 66.0 V**：设置输出过压保护
短按编码器移位，旋转编码器调整大小
 - 7. **S-OCP: 23.0 A**：设置输出过流保护
短按编码器移位，旋转编码器调整大小
 - 8. **S-OPP: 145 W**：设置输出过功率保护
短按编码器移位，旋转编码器调整大小
 - 9. **S-OTP: 060. °C**：设置过温保护
短按编码器移位，旋转编码器调整大小
 - 10. **S-OHP: ----: - h**：设置超时间保护
长按编码器打开/关闭
 - 11. **S-OAH: ----: - Ah**：设置超容量保护
长按编码器打开/关闭
 - 12. **S-OWH: ----: - Wh**：设置超能量保护
长按编码器打开/关闭
- S-OHP: 0000: 0 h** 在打开状态短按编码器移位，旋转编码器调整大小
- S-OAH: 0000. 00 Ah** 在打开状态短按编码器移位，旋转编码器调整大小
- S-OWH: 0000000. 0 Wh** 在打开状态短按编码器移位，旋转编码器调整大小



下载电源APP



下载上位机

下面附Modbus通讯协议

一. 协议简介

采用 TTL 串口传输接口，通信协议为 MODBUS-RTU 协议， 本产品只支持功能码 0x03、 0x06、 0x10。

二. 通信协议介绍

信息传输为异步方式， Modbus-RTU 模式以 11 位的字节为单位

起始位	1 位
数据位	8 位
奇偶校验位	无
停止位	1 位

数据帧结构：

数据帧间隔	地址码	功能码	数据区	CRC 校验
3.5 字节以上	1 字节	1 字节	N 字节	2 字节

发送数据前要求数据总线静止时间即无数据发送时间大于 3.5（例如：波特率为 9600 时为 5ms）消息发送至少要以 3.5 个字节时间的停顿间隔开始，整个消息帧必须作为一连 续的数据传输流，如果在帧完成之前有超过 3.5 个字节时间的停顿时间，接收设备将刷新 不完整的消息并假定下一字节是一个新消息的地址域。同样地，如果一个新消息在小于 3.5 个字符时间内接着前个消息开始，接收的设备将认为它是前一消息的延续。

1.1 地址码：

地址码是每次通讯信息帧的第一字节（8 位），从 1 到 255(初始默认为 1,0 同时也为广播地址)。这个字节表明由用户设置地址的从机将接收由主机发送来的信息。每个从机都必须有唯一的地址码，并且只有符合地 址码的从机才能响应回送信息。当从机回送信息时，回送数据均以各自的地址码开始。主机发送的地址码表明将发送到从机地址，而从机返回的地址码表明回送的从机地址。相应的地址码表明该信息来自于何处。

1.2 功能码：

功能码为每次通讯信息帧传送的第二个字节，ModBus 通讯规约可定义的功能码为 1 到 127。作为主机请求发送，通过功能码告诉从机应执行什么动作。作为从机响应，从机返回 的功能码与从主机发送来的功能码一样，并表明从机已响应主机并且已进行相关的操作。 本机仅支持 0x03、0x06、0x10 功能码。

功能码	定 义	操 作（二进制）
0x03	读寄存器数据	读取一个或多个寄存器的数据
0x06	写单个寄存器	把一组二进制数据写入单个寄存器
0x10	写多个寄存器	把多组二进制数据写入多个寄存器

1.3 数据区

数据区包括需要由从机返送何种信息或执行什么动作， 这些信息可以是数据（如：开 关量输入/输出、模拟量输入/输出、寄存器等等）、参考地址等。例如，主机通过功能码 03 告诉从机返回寄存器的值（包含要读取寄存器的起始地址及读取寄存器的长度），则返回的数据包括寄存器的数据长度及数据内容。（单个寄存器地址内的数据为双字节型数据）

0x03 读取功能主机格式

地址码	功能码	寄存器起始地址	寄存器地址数量 n(1～32)	CRC 校验码
1 字节	1 字节	2 字节	2 字节	2 字节
01	03	00 00	00 10	44 06

0x03 读取功能从机返回格式

地址码	功能码	返回寄存器个数 n	寄存器数据	CRC 校验码
1 字节	1 字节	1 字节	2*n 个字节	2 字节
01	03	04	2*n	

0x06 写单个寄存器功能主机格式

地址码	功能码	寄存器起始地址	寄存器数据	CRC 校验码
1 字节	1 字节	2 字节	2 字节	2 字节
01	06	00 12	00 01	E8 0F

0x06 写单个寄存器功能从机返回格式

地址码	功能码	寄存器起始地址	寄存器数据	CRC 校验码
1 字节	1 字节	2 字节	2 个字节	2 字节
01	06	00 12	00 01	E8 0F

0x10 写多个寄存器功能主机格式

地址码	功能码	寄存器起始地址	寄存器地址数量 n(1~32)	写入字节数 2*n	寄存器数据	CRC 校验码
1 字节	1 字节	2 字节	2 字节	1 字节	2*n 字节	2 字节

0x10 写多个寄存器从机主机格式

地址码	功能码	寄存器起始地址	寄存器地址数量 n(1~32)	CRC 校验码
1 字节	1 字节	2 字节	1 字节	2 字节

协议寄存器介绍(单个寄存器地址内的数据为双字节型数据)

名称	说明	字节数	小数点	单位	读写	寄存器地址
V-SET	电压设置	2	2	V	R/W	0000H
I-SET	电流设置	2	2	A	R/W	0001H
VOUT	输出电压显示值	2	2	V	R	0002H
IOUT	输出电流显示值	2	2	A	R	0003H
POWER	输出功率显示值	2	1	W	R	0004H
UIN	输入电压显示值	2	2	V	R	0005H
AH-LOW	输出 AH 低 16 位	2	0	maH	R	0006H
AH-HIGH	输出 AH 高 16 位	2	0	maH	R	0007H
WH-LOW	输出 WH 低 16 位	2	0	mwH	R	0008H
WH-HIGH	输出 WH 高 16 位	2	0	mwH	R	0009H
OUT_H	开启时长-小时	2	0	H	R	000AH
OUT_M	开始时长-分钟	2	0	M	R	000BH
OUT_S	开启时长-秒	2	0	S	R	000CH
T_IN	内部温度值	2	1	C	R	000DH
T_EX	外部温度值	2	1	C	R	000EH
LOCK	按键锁	2	0	-	R/W	000FH
PROTECT	保护状态	2	0	-	R/W	0010H

CVCC	恒压恒流状态	2	0	—	R	0011H
ONOFF	开关输出	2	0	—	R/W	0012H
F-C	温度符号	2	0	—	R	0013H
B-LED	背光亮度等级	2	0	—	R/W	0014H
SLEEP	息屏时间	2	0	M	R/W	0015H
MODEL	产品型号	2	0	—	R	0016H
VERSION	固件版本号	2	0	—	R	0017H
SLAVE-ADD	从机地址	2	0	—	R/W	0018H
BAUDRATE_L	波特率	2	0	—	R/W	0019H
T-IN-OFFSET	内部温度修正	2	1	C	R/W	001AH
T-EX-OFFSET	外部温度修正	2	1	C	R/W	001BH
BUZZER	蜂鸣器开关	2	0	—	R/W	001CH
EXTRACT-M	快捷调出数据组	2	0	—	R/W	001DH
DEVICE	设备状态	2	0	—	R/W	001EH
V-SET	电压设置	2	2	V	R/W	001FH
V-SET	电压设置	2	2	V	R/W	0050H
I-SET	电流设置	2	2	A	R/W	0051H
S-LVP	低压保护值	2	2	V	R/W	0052H
S-OVP	过压保护值	2	2	V	R/W	0053H
S-OCP	过流保护值	2	2	A	R/W	0054H
S-OPP	过功率保护值	2	1	W	R/W	0055H
S-OHP_H	最大输出时长一小时	2	0	H	R/W	0056H
S-OHP_M	最大输出时长一分钟	2	0	M	R/W	0057H
S-OAH_L	最大输出 AH 低 16 位	2	0	maH	R/W	0058H
S-OAH_H	最大输出 AH 高 16 位	2	0	maH	R/W	0059H
S-OWH_L	最大输出 WH 低 16 位	2	0	lOmwH	R/W	005AH
S-OWH_H	最大输出 WH 高 16 位	2	0	lOmwH	R/W	005BH
S-OTP	过温保护值	2	1	F/C	R/W	005CH
S-INI	上电输出开关	2	0	—	R/W	005DH
S-0_CLOSE	过时 过容量 过能量开关 最后一位 开关状态 0000000000000001 过时 0000000000000010 过容 0000000000000100 过能 0000000100000000	2	0	—	R/W	005EH
Restore Factory	恢复出厂设置	2	0		W	0020H
ZERO	清零	2	0	—	W	0021H
Reset	重启	2	0		W	002FH

注 1:本产品设计与 M0-M10 共 11 组存储数据组，每组有序号 00-0D 共 14 个数据，其中 M0 数据组为产品上电默认调用的数据组，数据组的起始地址计算方法是： 0050H+数据组号*0010H, 例如 M3 数据组的起始地址为：0050H+3*0010H=0080H。

注 2:按键锁功能读写数值为 0 和 1， 0 为非锁定， 1 为锁定。

注 3:保护状态读取值为 0-11：
0:正常运行, 1:OVP, 2:OCP, 3:OPP, 4:LVP, 5:OAH, 6:OHP, 7:OTP, 8:OEP, 9:OWH, 10:ICP。 11： IVP

注 4:恒压恒流状态读取值为 0 和 1， 0 为 CV 状态， 1 为 CC 状态。

注 5:开关输出功能读写值为 0 和 1， 0 为关闭状态， 1 为打开状态。

注 6:背光亮度等级读写范围为 0-5， 0 级最暗， 5 级最亮。

注 7:快捷调出数据组功能写入值为 0-10， 写入后会自动调出对应数据组数据。

1.4 错误校验码（CRC 校验）：
主机或从机可用校验码进行判别接收信息是否正确。由于电子噪声或一些其它干扰，信息在传输过程中有时会发生错误，错误校验码（CRC）可以检验主机或从机在通讯数据传送过程中的信息是否有误，错误的信息可以放弃（无论是发送还是接收），这样增加了系统的安全和效率。 MODBUS 通讯协议的 CRC（冗余循环码）包含 2 个字节，即 16 位二进制数。CRC 码由发送设备（主机）计算，放置于发送信息帧的尾部。接收信息的设备（从机）再重新计算接收到信息的 CRC，比较计算得到的 CRC 是否与接收到的相符，如果两者不相符，则表明出错。 CRC 校验码发送时低位在前，高位在后。

CRC 码的计算方法：
(1) 预置 1 个 16 位的寄存器为十六进制 FFFF（即全为 1）；称此寄存器为 CRC 寄存器；
(2) 把第一个 8 位二进制数据（既通讯信息帧的第一个字节）与 16 位的 CRC 寄存器的低 8 位相异或，把结果放于 CRC 寄存器；
(3) 把 CRC 寄存器的内容右移一位（朝低位）用 0 填补最高位，并检查右移后的移出位；
(4) 如果移出位为 0：重复第 3 步（再次右移一位）；如果移出位为 1： CRC 寄存器与多项式 A001（1010 0000 0000 0001）进行异或；
(5) 重复步骤 3 和 4，直到右移 8 次，这样整个 8 位数据全部进行了处理；
(6) 重复步骤 2 到步骤 5，进行通讯信息帧下一个字节的处理；
(7) 将该通讯信息帧所有字节按上述步骤计算完成后，得到的 16 位 CRC 寄存器的高、低字节进行交换；
(8) 最后得到的 CRC 寄存器内容即为 CRC 码。

三、通讯实例

例 1： 主机读取输出电压和输出电流显示值			
主机发送的报文格式：			
主机发送	字节数	发送的信息	备 注
从机地址	1	01	发送至地址为 01 的从机
功能码	1	03	读寄存器
寄存器起始地址	2	0002H	寄存器起始地址
寄存器地址数量	2	0002H	共 2 个字节
CRC 码	2	65CBH	由主机计算得到 CRC 码

例如如当前显示值是 05.00V， 2.00A，则从机响应返回的报文格式：			
从机响应	字节数	返回的信息	备 注
从机地址	1	01	来自从机 01
功能码	1	03	读寄存器
读取字节数	1	04	共 1 个字节
地址为 0002H 寄存器的内容	2	01F4H	输出电压显示值
地址为 0003H 寄存器的内容	2	00C8H	输出电流显示值

CRC 码	2	BBABH	由从机计算得到 CRC 码
-------	---	-------	---------------

例 2： 主机要设定电压为 12. 00V

主机发送的报文格式：

主机发送	字节数	发送的信息	备 注
从机地址	1	01H	来自从机 01
功能码	1	06H	写单个寄存器
寄存器地址	2	0000H	寄存器地址
地址为 0000H 寄存器的内容	2	04B0H	设定输出电压值
CRC 码	2	8ABEH	由主机计算得到 CRC 码

从机接收后响应返回的报文格式：

从机响应	字节数	返回的信息	备 注
从机地址	1	01H	发送至地址为 01 的从机
功能码	1	06H	写单个寄存器
寄存器地址	2	0000H	寄存器起始地址
地址为 0000H 寄存器的内容	2	04B0H	设定输出电压值
CRC 码	2	8ABEH	由从机计算得到 CRC 码

例 3：主机要设定电压为 24. 00V，电流 15. 00A。

主机发送的报文格式：

主机发送	字节数	发送的信息	备 注
从机地址	1	01H	来自从机 01
功能码	1	10H	写寄存器
寄存器起始地址	2	0000H	寄存器起始地址
寄存器地址数量	2	0002H	共 2 个字节
写入字节数	1	04H	共 1 个字节
地址为 0000H 寄存器的内容	2	0960H	设定输出电压值
地址为 0001H 寄存器的内容	2	05DCH	设定输出电流值
CRC 码	2	F2E4H	由主机计算得到 CRC 码

从机接收后响应返回的报文格式：

从机响应	字节数	返回的信息	备 注
从机地址	1	01H	发送至地址为 01 的从机
功能码	1	10H	写寄存器
寄存器起始地址	2	0000H	寄存器起始地址
寄存器地址数量	2	0002H	共 2 个字节
CRC 码	2	41C8H	由从机计算得到 CRC 码

Third generation upgraded CNC voltage buck power supply color screen

Model: ZK-6522C

Product parameters

Product name: CNC DC buck power supply (Color screen) Product model: ZK-6522C
Input voltage: 6.0-75.00V Output voltage: 0.0-65.00V
Output current: 0-22A Output voltage accuracy: $\pm 0.3\%+3$ words (Calibratable)
Output power: 0-1430W; Output current accuracy: $\pm 0.5\%+3$ words (Calibratable)
Voltage resolution: 0.01V Current resolution: 0.01A
Data group storage: 11 groups Silicon rubber keypads: 5
Screen size: Upgraded 1.8-inch large screen, 36 * 29mm visible range Buzzer: Yes
Cooling fan on conditions: Temperature > 45 °C or current > 2A or power > 30W
Product size: Product shell 83x48x29.4mm; (Height does not include rotary encoder)
Product motherboard: 124x71x42mm
Product weight: Net weight of motherboard 210g, Net weight of the entire set 243g, The entire set includes a packaging weight of 296g

Protection mechanism:

Input undervoltage protection (LUP): 5.5-64V adjustable, factory default value of 5.5V
Output overvoltage protection (OUP): 1-66V adjustable, factory default value 66V
Output overcurrent protection (OCP): 0.01-23A adjustable, factory default value 23A
Output overpower protection (OPP): 1-1450W adjustable, factory default value 1450W
Over temperature protection (OTP): 30-99 °C adjustable, factory default value of 60°C
Timeout protection (OHP): 1-99 hours and 59 minutes, factory default off
Overcapacity protection (OAH): 0.001-9999Ah, factory default off
Superenergy protection (OWH): 0.001-4000KWh, factory default off

Notes :

1. It is not allowed to input AC power, let alone connect to 220V mains power.
2. The maximum allowable input voltage is 75V, and overvoltage input or reverse polarity connection is not allowed.
3. The output cannot fully guarantee anti backflow. When connecting the battery, it is necessary to ensure that the output voltage of the power module is always higher than the battery voltage, or to connect a diode in series with a unidirectional conduction at the output positive pole to prevent battery voltage backflow from burning out the power supply.
4. When connecting the output terminal to the battery, be careful not to reverse the positive and negative poles.
5. The product is a voltage reduction module, maximum output voltage = the input voltage * 0.95-1V.

Key functions

Short press: Call up data group settings

Short press: Voltage setting

Long press: OVP setting

Short press: Current setting

Long press: OCP setting

Short press: OK

Short press: Select setting bit

Long press: Open/Cancel lock function

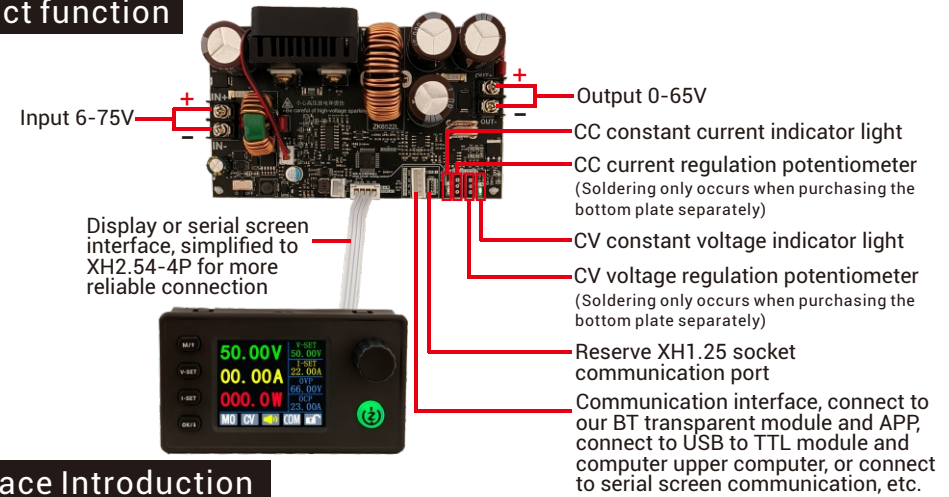
Left rotation: Switching interface display;
Number reduction during setup

Right rotation: Switching interface display;
Number increase during setup

Short press: Turn off/on output

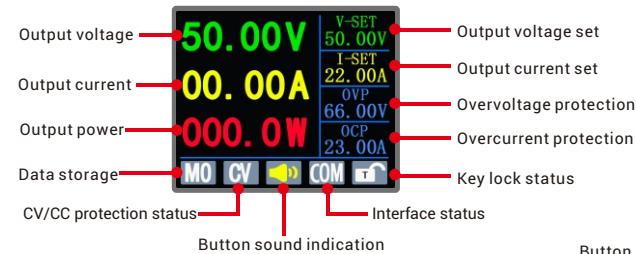
Long press: Sleep shutdown

Product function

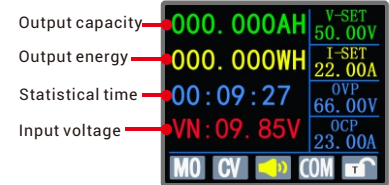


Interface Introduction

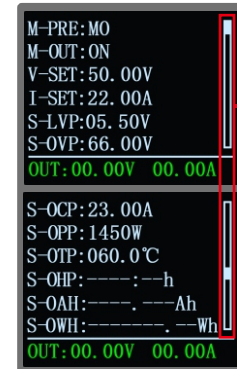
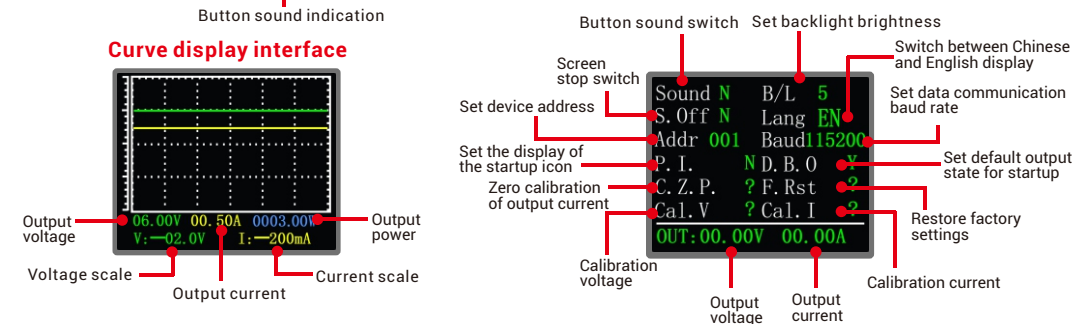
Main interface of power supply



Statistics Interface



Menu interface



Data group setting interface

M-PRE: Select data group M0-M10
M-OUT: Set the default switch when calling out the corresponding data group
V-SET: Set the output voltage value for the corresponding data group
I-SET: Set the output current value for the corresponding data group
S-LVP: Set the input undervoltage protection value for the corresponding data group, with a default of 5.5V
S-OVP: Set the output overvoltage protection value for the corresponding data group, default to 66V
S-OCP: Set the output overcurrent protection value for the corresponding data group, with a default of 23A
S-OPP: Set the output overpower protection value for the corresponding data group, with a default of 1450W
S-OTP: Set the over temperature protection value for the corresponding data group, default to 60°C
S-OHP: Set the timeout protection time for the corresponding data group, which is disabled by default
S-OAH: Set the over capacity protection value for the corresponding data group, which is disabled by default

Shell size



Suggested opening size

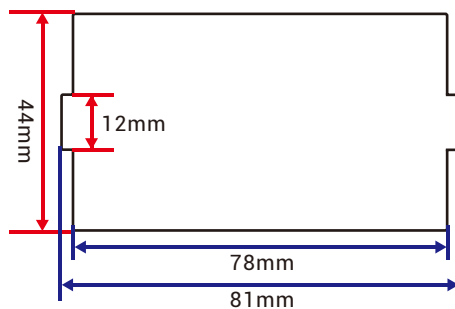
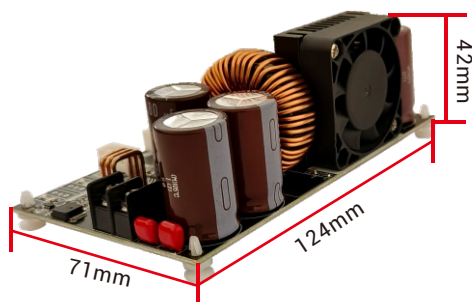
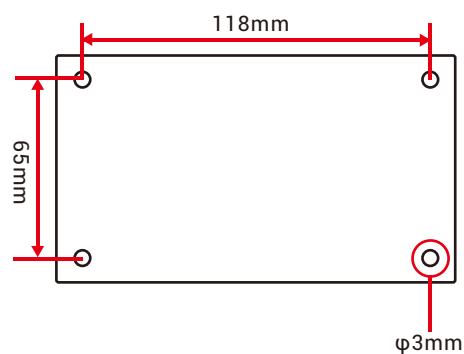


Plate size

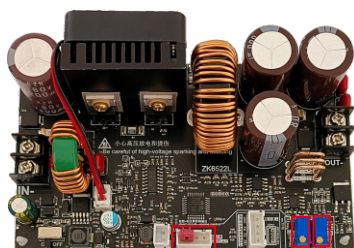


Bottom plate installation hole size



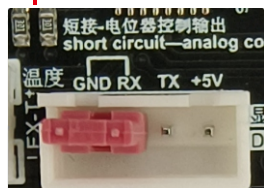
Independent use of motherboard

Output current 0-22A adjustable, output voltage 0-65V adjustable



Tip: When selling the ZK6522B motherboard separately, it is necessary to solder the voltage and current adjustment potentiometers and plug in the jumper cap. If it is a serial screen or APP/Upper computer control, the jumper cap must be unplugged and powered on again..

Short circuit RX GND is required for analog potentiometer control, otherwise digital communication control is required



Voltage regulation potentiometer
Current regulation potentiometer



APP online method



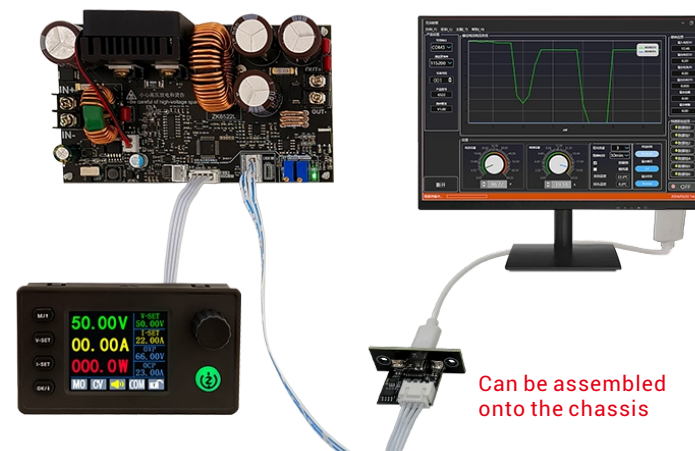
① Insert the communication interface of the power motherboard into the ZK-BT BT board, and the ZK-BT indicator light will flash once when inserted.

② Open the Wuzhi Zhilian APP and mobile BT, click "Add Device" on the APP interface, and Wuzhi Power will be automatically searched for. After clicking, it will automatically connect.

③ Display successful connection, click complete.

④ Click on the device icon to enter the normal operation interface. Long press the product icon to change the name or delete the device.

Online method of upper computer



Can be assembled onto the chassis

① Insert a data cable into the computer and connect it to the ZK-U2T (USB to TTL module). The other end of the ZK-U2T module has a 4P cable inserted into the communication port of the power motherboard;

② Find the port number COMx corresponding to USB-Serial CH340 in my computer - Device Manager - Ports;

③ Open the "Wuzhi Zhilian" upper computer, select the corresponding port number COMx, and click the "Connect" button in the bottom left corner of the upper computer to automatically connect.

④ The operating environment is If NET Framework 4.8 cannot run, you can download and install it from the .net official website

Instructions for use

Call out data group



Press **M/T** to call up the data group settings, rotate the encoder to switch to data group M0-M10;
Press **OK/↓** to confirm and exit

Set voltage and current



Short press **V-SET** to enter the voltage setting, short press to rotate the encoder to shift and adjust the size.
After setting up, press **V-SET** or **OK/↓** again to exit the setup.



Short press **I-SET** to enter the current setting, short press to rotate the encoder to shift and adjust the size.
After setting up, press **I-SET** or **OK/↓** again to exit the setup.

Quick settings OVP/OCF



Long press **V-SET** to enter OVP settings, short press the rotary encoder to shift, rotate to adjust size.
After setting up, press **V-SET** or **OK/↓** again to exit the setup.

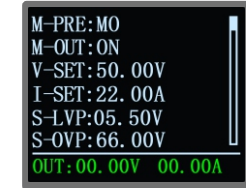
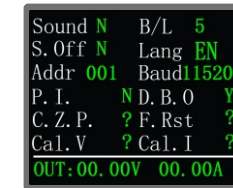
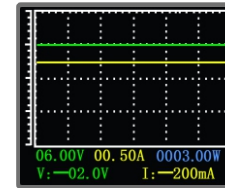


Long press **I-SET** to enter OCF settings, short press the rotary encoder to shift, rotate to adjust size.
After setting up, press **I-SET** or **OK/↓** again to exit the setup.

Switch display interface



Rotation encoder switch interface display



Locking function



Show lock symbols

On the normal display interface, long press and hold the rotary encoder for 2 seconds to lock the set voltage and current to prevent accidental operation; After locking, press and hold the encoder for 2 seconds to unlock.
After locking, the power button can still operate normally, ensuring that the power can be disconnected at any time.

Introduction to parameter setting operations

On the normal interface, the rotary encoder switch interface displays

Curve display interface



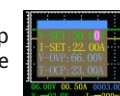
Press **OK/↓** to set the voltage scale (display **V: -02.0V**), rotate the encoder to switch size, then press **OK/↓** to switch to the current scale (display **I: -200mA**), rotate the encoder to switch size, and then press **OK/↓** to exit; Short press the rotary encoder to pause.

Press **M/T** to call out the data group



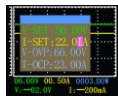
Rotating encoder switching data group M0-M10;
Press **OK/↓** to confirm and exit

Press **V-SET** to bring up the settings interface



Can set voltage, current, overvoltage protection, overcurrent protection

Press **I-SET** to bring up the settings interface



Can set voltage, current, overvoltage protection, overcurrent protection

Menu interface



Short press **M/t** to select the settings option

1. **Sound N** : Rotating encoder on/off
2. **S. Off N** : Turn the rotary encoder on/off and set the screen rest time
3. **Addr 001** : Set communication address for rotary encoder
4. **P. L N** : Rotating encoder on/off

5. **C. Z. P. ?** : Short press **OK/↓** to pop up the calibration zero point dialog box



Short press **M/t** to cancel and exit, Short press **OK/↓** to confirm and exit.

6. **Cal. V ?** : Short press **OK/↓** to pop up the calibration voltage dialog box



Short press **M/t** in the dialog box to cancel calibration and exit.

Calibration steps:

1. Use a multimeter to measure the actual voltage at the current output terminal (output terminal unloaded, measure the power terminal) and input it.
2. Short press **OK/↓** to display voltage 2.
3. Use a multimeter to measure the actual voltage at the current output terminal again and input it.
4. Short press **OK/↓** to complete the calibration.
If calibration is successful, it will display "CALI Success" and exit; otherwise, it will display "CALI Failed" and exit. If it fails, you can try calibrating again.

7. **B/L 5** : Rotate the encoder to adjust the backlight brightness; 0-5 levels
8. **Lang EN** : Rotating encoder switching between Chinese and English
9. **Baud 115200** : Rotation encoder switching communication rate size
10. **D.B.O Y** : Rotate the encoder to adjust the default output on/off during startup
11. **F. Rst ?** : Short press **OK/↓** to pop up the calibration and factory reset dialog box



Short press **M/t** to cancel and exit, Short press **OK/↓** to confirm and exit.

12. **Cal. I ?** : Short press **OK/↓** to pop up the calibration current dialog box



Short press **M/t** in the dialog box to cancel calibration and exit.

Calibration steps:

1. Use a multimeter current range or electronic load to short circuit the output terminals and measure the current actual current and input it.
2. Short press **OK/↓** to display current 2.
3. Use a multimeter to measure the actual current at the current output terminal again and input it.
4. Short press **OK/↓** to complete the calibration.
If calibration is successful, it will display "CALI Success" and exit; otherwise, it will display "CALI Failed" and exit. If it fails, you can try calibrating again.

Data group setting interface

Short press **M/t** to select the settings option, and once the settings are complete, short press **OK/↓** to confirm and exit.



1. **M-PRE: M0** : **Set up data groups**
Rotating encoder selection data group
2. **M-OUT: ON** : **Set default on/off when calling out**
Default on/off when calling up rotary encoder settings
3. **V-SET: 50.0 V** : **Set output voltage**
Short press the encoder to shift, rotate the encoder to adjust size
4. **I-SET: 22.0 A** : **Set output current**
Short press the encoder to shift, rotate the encoder to adjust size
5. **S-LVP: 05.5 V** : **Set input undervoltage protection**
Short press the encoder to shift, rotate the encoder to adjust size
6. **S-OVP: 66.0 V** : **Set output overvoltage protection**
Short press the encoder to shift, rotate the encoder to adjust size
7. **S-OCP: 23.0 A** : **Set output overcurrent protection**
Short press the encoder to shift, rotate the encoder to adjust size
8. **S-OPP: 145 W** : **Set output overpower protection**
Short press the encoder to shift, rotate the encoder to adjust size
9. **S-OTP: 060. °C** : **Set up over temperature protection**
Short press the encoder to shift, rotate the encoder to adjust size
10. **S-OHP: ----. --h** : **Set timeout protection** **S-OHP: 0000.0 h** Short press the encoder shift in the open state, rotate the encoder to adjust size
Long press the encoder to turn on/off
11. **S-OAH: ----. --Ah** : **Set up over capacity protection** **S-OAH: 0000.00 Ah** Short press the encoder shift in the open state, rotate the encoder to adjust size
Long press the encoder to turn on/off
12. **S-OWH: ----. --Wh** : **Set up over energy protection** **S-OWH: 0000000.0 Wh** Short press the encoder shift in the open state, rotate the encoder to adjust size
Long press the encoder to turn on/off



Download the power app



Download the upper computer

Attached below is the Modbus communication protocol

一. Protocol Introduction

Using TTL serial transmission interface and MODBUS-RTU communication protocol, this product only supports function codes 0x03, 0x06, and 0x10.

2、 Introduction to Communication Protocol:

Information transmission is asynchronous, and Modbus RTU mode is based on 11 bit bytes

Starting position	1 digit
Data bits	8 digits
Parity check bit	nothing
Stop bit	1 digit

Data frame structure:

Data frame interval	Address code	Function code	Data area	CRC verification
3.5 bytes or more	1 byte	1 byte	N bytes	2 bytes

Before sending data, it is required that the data bus rest time, i.e. no data transmission time, be greater than 3.5 (for example, 5ms at a baud rate of 9600). Message transmission must start with a pause interval of at least 3.5 bytes, and the entire message frame must be treated as a continuous data transmission stream. If there is a pause time of more than 3.5 bytes before the frame is completed, the receiving device will refresh the incomplete message and assume that the next byte is the address domain of a new message. Similarly, if a new message starts with the previous message within less than 3.5 characters, the receiving device will consider it a continuation of the previous message.

1. 1 Address Code:

The address code is the first byte (8 bits) of each communication information frame, ranging from 1 to 255 (initially set to 1,0 and also the broadcast address). This byte indicates that the slave set by the user will receive information sent by the host. Each slave must have a unique address code, and only slaves that match the address code can respond to feedback messages. When the slave sends back information, the returned data starts with their respective address codes. The address code sent by the host indicates the slave address to be sent, while the address code returned by the slave indicates the slave address to be returned. The corresponding address code indicates where the information comes from.

1. 2 Function code:

The function code is the second byte transmitted in each communication information frame, and the ModBus communication protocol can define function codes ranging from 1 to 127. Sent as a host request, telling the slave what action to take through a function code. As a response from the slave, the function code returned by the slave is the same as the function code sent from the master, and indicates that the slave has responded to the master and performed relevant operations. This machine only supports function codes 0x03, 0x06, and 0x10.

Function code	Definition	Operation (binary)
0x03	Read register data	Read data from one or more registers
0x06	Write a single register	Write a set of binary data into a single register
0x10	Write multiple registers	Write multiple sets of binary data into multiple registers

1. 3 Data Area

The data area includes what information or actions need to be returned by the slave, which can be

data (such as switch inputs/outputs, analog inputs/outputs, registers, etc.), reference addresses, etc. For example, if the host tells the slave through function code 03 to return the value of the register (including the starting address of the register to be read and the length of the read register), the returned data includes the length and content of the register. (The data within a single register address is double byte data)

0x03 Read function host format

Address code	Function code	Register Start Address	Number of register addresses n (1-32)	CRC verification code
1 byte	1 byte	2 bytes	2 bytes	2 bytes
01	03	00 00	00 10	44 06

0x03 Read Function Slave Return Format

Address code	Function code	Return the number of registers n	Register data	CRC verification code
1 byte	1 byte	1 byte	2*n 个字节	2 bytes
01	03	04	2*n	

0x06 Write Single Register Function Host Format

Address code	Function code	Register Start Address	Register data	CRC verification code
1 byte	1 byte	2 bytes	2 bytes	2 bytes
01	06	00 12	00 01	E8 0F

0x06 Write Single Register Function Slave Return Format

Address code	Function code	Register Start Address	Register data	CRC verification code
1 byte	1 byte	2 bytes	2 bytes	2 bytes
01	06	00 12	00 01	E8 0F

0x10 Write Multiple Registers Function Host Format

Address code	Function code	Register Start Address	Number of register addresses n (1-32)	Write Bytes 2*n	Register data	CRC verification code
1 byte	1 byte	2 bytes	2 bytes	1 byte	2*n 字节	2 bytes

0x10 Write multiple registers in slave host format

Address code	Function code	Register Start Address	Number of register addresses n (1-32)	CRC verification code
1 byte	1 byte	2 bytes	1 byte	2 bytes

Introduction to Protocol Registers (Data within a Single Register Address is Double Byte Data)

name	explain	Byte count	decimal point	unit	Reading and	Register Address
------	---------	------------	---------------	------	-------------	------------------

					writing	
V-SET	Voltage setting	2	2	V	R/W	0000H
I-SET	Current setting	2	2	A	R/W	0001H
VOUT	Output voltage display value	2	2	V	R	0002H
IOUT	Output current display value	2	2	A	R	0003H
POWER	Output power display value	2	1	W	R	0004H
UIN	Input voltage display value	2	2	V	R	0005H
AH-LOW	Output AH low 16 bits	2	0	maH	R	0006H
AH-HIGH	Output AH high 16 bits	2	0	maH	R	0007H
WH-LOW	Output WH low 16 bits	2	0	mwH	R	0008H
WH-HIGH	Output WH high 16 bits	2	0	mwH	R	0009H
OUT-H	Opening duration - hours	2	0	H	R	000AH
OUT-M	Start time in minutes	2	0	M	R	000BH
OUT-S	Opening time - seconds	2	0	S	R	000CH
T. IN	Internal temperature value	2	1	C	R	000DH
T. EX	External temperature value	2	1	C	R	000EH
LOCK	Key lock	2	0	-	R/W	000FH
PROTECT	Protection status	2	0	-	R/W	0010H
CVCC	Constant voltage and constant current state	2	0	-	R	0011H
ONOFF	Switch output	2	0	-	R/W	0012H
F-C	Temperature symbol	2	0	-	R	0013H
B-LED	Backlight brightness level	2	0	-	R/W	0014H
SLEEP	Screen rest time	2	0	M	R/W	0015H
MODEL	Product model	2	0	-	R	0016H
VERSION	Firmware Version	2	0	-	R	0017H
SLAVE-ADD	Slave address	2	0	-	R/W	0018H
BAUDRATE-L	Baud rate	2	0	-	R/W	0019H
T-IN-OFFSET	Internal temperature correction	2	1	C	R/W	001AH
T-EX-OFFSET	External temperature correction	2	1	C	R/W	001BH
BUZZER	Buzzer switch	2	0	-	R/W	001CH
EXTRACT-M	Quick access to data groups	2	0	-	R/W	001DH
DEVICE	Device status	2	0	-	R/W	001EH
V-SET	Voltage setting	2	2	V	R/W	001FH
V-SET	Voltage setting	2	2	V	R/W	0050H
I-SET	Current setting	2	2	A	R/W	0051H

S-LVP	Low voltage protection value	2	2	V	R/W	0052H
S-OVP	Overvoltage protection value	2	2	V	R/W	0053H
S-OCp	Overcurrent protection value	2	2	A	R/W	0054H
S-OPP	Overpower protection value	2	1	W	R/W	0055H
S-OHP_H	Maximum output duration - hours	2	0	H	R/W	0056H
S-OHP_M	Maximum output duration in minutes	2	0	M	R/W	0057H
S-OAH.L	Maximum output AH low 16 bits	2	0	maH	R/W	0058H
S-OAH_H	Maximum output AH high 16 bits	2	0	maH	R/W	0059H
S-OWH.L	Maximum output WH low 16 bits	2	0	10mwH	R/W	005AH
S-OWH.H	Maximum output WH high 16 bits	2	0	10mwH	R/W	005BH
S-OTP	Over temperature protection value	2	1	F/C	R/W	005CH
S-INI	Power on output switch	2	0	-	R/W	005DH
S-O-CLOSE	The last switch state of the outdated over capacity and over energy switch Obsolete Over capacity 10000000000000 100 Overpower 000000 10000000	2	0	-	R/W	005EH
Restore Factory	Restore factory settings	2	0		W	0020H
ZERO	Zeroing	2	0	-	W	0021H
Reset	restart	2	0		W	002FH

Note 1: This product is designed with 11 storage data groups of M0-M10, each with a total of 14 data groups numbered 00-0D. The M0 data group is the default data group called by the product when powered on, and the starting address calculation method for the data group is: 0050H+data group number * 0010H. For example, the starting address for the M3 data group is 0050H+3 * 0010H=0080H.

Note 2: The reading and writing values for the key lock function are 0 and 1, where 0 is non locked and 1 is locked.

Note 3: The read value for protection status is 0-11:

0: Normal operation, 1: OVP, 2: OCP, 3: OPP, 4: LVP, 5: OAH, 6: OHP, 7: OTP, 8: OEP, 9: OWH, 10: ICP.

11: IVP

Note 4: The reading values for constant voltage and current state are 0 and 1, where 0 is the CV state and 1 is the CC state.

Note 5: The read and write values for the switch output function are 0 and 1, with 0 being the off state and 1 being the on state.

Note 6: The reading and writing range of the backlight brightness level is 0-5, with level 0 being the darkest and level 5 being the brightest.

Note 7: The quick access data group function writes values from 0 to 10, and after writing, the corresponding data group data will be automatically retrieved.

1.4 Error Check Code (CRC Check):

The host or slave can use a checksum to determine whether the received information is correct. Due to electronic noise or other interferences, errors may sometimes occur during the transmission of information. Error check codes (CRC) can verify whether the information of the host or slave during communication data transmission is correct. Incorrect data can be discarded (whether sent or received), which increases the security and efficiency of the system. The CRC (Redundant Cyclic Code) of MODBUS communication protocol contains 2 bytes, which is a 16 bit binary number. The CRC code is calculated by the transmitting device (host) and placed at the end of the transmitted information frame. The receiving device (slave) recalculates the CRC of the received information and compares whether the calculated CRC matches the received CRC. If the two do not match, it indicates an error. When sending CRC checksum, the low bit is in front and the high bit is in the back.

Calculation method for CRC code:

- (1) Set one 16 bit register as hexadecimal FFFF (i.e. all are 1); Call this register CRC register;
- (2) Distinguish the first 8-bit binary data (i.e. the first byte of the communication information frame) from the lower 8 bits of the 16 bit CRC register, and place the result in the CRC register;
- (3) Shift the content of the CRC register to the right by one bit (towards the low bit) and fill the highest bit with 0, and check the shifted out bit after the right shift;
- (4) If the displacement is 0: repeat step 3 (move one position to the right again); If the removal bit is 1: CRC register XOR with polynomial A001 (1010 0000 0000 0001);
- (5) Repeat steps 3 and 4 until shifted 8 times to the right, so that the entire 8-bit data has been processed;
- (6) Repeat steps 2 to 5 to process the next byte of the communication information frame;
- (7) After calculating all bytes of the communication information frame according to the above steps, swap the high and low bytes of the 16 bit CRC register obtained;
- (8) The final CRC register obtained is the CRC code.

3、Communication

Example 1: The host reads the displayed values of output voltage and output current

Message format sent by the host:

Host sending	Byte count	Message sent	Remarks
Slave address	1	01	Sent to slave with address 01
Function code	1	03	Read Register
Register Start Address	2	0002H	Register Start Address
Number of register addresses	2	0002H	2 bytes in total
CRC code	2	65CBH	CRC code calculated by the host

For example, if the current displayed value is 05.00V, 2.00A, the format of the message returned by the slave response is:

Slave response	Byte count	Information returned	Remarks
Slave address	1	01	From slave 01
Function code	1	03	Read Register
Read Bytes	1	04	1 byte in total
The content of register with address 0002H	2	01F4H	Output voltage display value
The content of register with address 0003H	2	00C8H	Output current display value
CRC code	2	BBABH	CRC code calculated by slave machine

Example 2: The host needs to set the voltage to 12.00V

Message format sent by the host:

Host sending	Byte count	Message sent	Remarks
Slave address	1	01H	From slave 01
Function code	1	06H	Write a single register
Register Address	2	0000H	Register Address
The content of register with address 0000H	2	04B0H	Set output voltage value
CRC code	2	8ABEH	CRC code calculated by the host

The format of the message returned by the response received by the slave is:

Slave response	Byte count	Information returned	Remarks
Slave address	1	01H	Sent to slave with address 01
Function code	1	06H	Write a single register
Register Address	2	0000H	Register Start Address
The content of register with address 0000H	2	04B0H	Set output voltage value
CRC code	2	8ABEH	CRC code calculated by slave machine

Example 3: The host needs to set the voltage to 24.00V and the current to 15.00A.

Message format sent by the host:

Host sending	Byte count	Message sent	Remarks
Slave address	1	01H	From slave 01
Function code	1	10H	Write Register
Register Start Address	2	0000H	Register Start Address
Number of register addresses	2	0002H	2 bytes in total
Write Bytes	1	04H	1 byte in total
The content of register with address 0000H	2	0960H	Set output voltage value
The content of register 0001H with address	2	05DCH	Set output current value

CRC code	2	F2E4H	CRC code calculated by the host
----------	---	-------	---------------------------------

The format of the message returned by the response received by the slave is:

Slave response	Byte count	Information returned	Remarks
Slave address	1	01H	Sent to slave with address 01
Function code	1	10H	Write Register
Register Start Address	2	0000H	Register Start Address
Number of register addresses	2	0002H	2 bytes in total
CRC code	2	41C8H	CRC code calculated by slave machine